Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

Impact Assessment and Mitigation of Electric Vehicle Smart Charging Harmonics

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This work was supported by the Turkish Ministry of National Education.

ABSTRACT Smart charging of electric vehicles (EVs) offers significant opportunities for grid operators and EV users, including deferring costly network reinforcements and reducing charging costs. However, reduced charging rates inherently introduce higher harmonic content, posing a serious risk to distribution transformers in heavily loaded networks and elevating total harmonic distortion (THD) levels in mildly loaded ones. This paper leverages harmonic profiles from eight EV models to quantify the impact of large-scale EV charging on both heavily and mildly loaded distribution systems. For heavily loaded networks, a Monte Carlo-based framework is developed to evaluate transformer aging under various charging current scenarios, and a rule-based harmonics-averse EV charging management strategy is proposed for peak demands. For mildly loaded networks, a harmonics-aware smart charging scheme is introduced to ensure THD remains within regulatory limits. The underlying optimization problem is solved using particle swarm optimization, augmented with a Water Filling algorithm to address discontinuities in infeasible charging rates (e.g., 0–1.38 kW). Numerical results demonstrate that by strategically adjusting charging rates, THD compliance can be achieved without costly network upgrades.

INDEX TERMS Electric vehicles, harmonic emissions, smart charging, transformer aging

I. INTRODUCTION

THE global drive toward decarbonizing transportation is centered on the large-scale adoption of electric vehicles (EVs), as road transport remains one of the dominant sources of greenhouse gas (GHG) emissions. For instance, road vehicles account for approximately 22% of total US GHG emissions [1], while in the UK, transport contributed about 28% of national GHG emissions in 2022, making it the largest emitting sector [2]. Similarly, transport represents nearly one-quarter of the EU's total GHG emissions, with road transport constituting about 75% of these emissions [3]. Over the last decade, the number of EVs on the road has surged from a few thousand to millions worldwide, supported by aggressive policy measures, with EVs accounting for over 20% of new vehicle sales in 2023 [4]. As of early 2025, the UK has over 1.4 million pure EVs, representing more than 4% of the national vehicle fleet [5], while countries like Norway reached over 90% EV share in new vehicle sales in 2024 [6].

Despite these advances, the sustained growth of EV adoption requires significant investments in charging infrastructure and grid reinforcements to provide sufficient coverage and capacity [7]. However, such infrastructure upgrades involve multiple stakeholders and lengthy approval processes (e.g., [8] outlines typical UK timelines). To address capacity constraints and defer costly investments, smart charging has emerged as an effective EV load management strategy [9]. By dynamically coordinating EV charging schedules and rates with real-time grid conditions, smart charging can mitigate transformer overloading, reduce peak demand, and enhance system reliability [10]. In practice, smart charging mechanisms allow controllers to: (i) adjust charging currents (e.g., 6–16 A for most EVs); (ii) turn charging sessions on or off; (iii) shift between phases; and (iv) switch between single-phase and three-phase charging configurations [11].

In recognition of these benefits, smart charging is now being mandated in various regions, including the EU and UK, where charging points are required to support remote monitoring and control for energy optimization purposes. In the EU, smart charging is regulated under the Alternative Fuels Infrastructure Regulation (AFIR) [12], and in the UK by the Electric Vehicles (Smart Charge Points) Regulations [13]. Beyond technical grid benefits, smart charging also offers financial incentives for consumers through time-of-use pricing and demand shifting [14], [11]. However, despite these advan-

tages, increasing attention is being paid to the impact of smart charging on EV on-board charger performance [15]. Since in AC charging, the EV load profile is primarily governed by the on-board charger [16], key performance metrics such as power factor, efficiency, and harmonic distortion become critical for both grid and asset health. Recent studies indicate that harmonic behaviour significantly varies with flexible charging schedules and different charger topologies, highlighting that harmonics are critical in future EV integration frameworks [17], [18], [19]. This underscores the need for incorporating harmonics into smart charging decisions rather than treating them as a static compliance assessment.

A. RESEARCH GAPS

While extensive research exists on smart charging and its implications for network assets [20], recent studies highlight that varying charging rates introduce substantial harmonic content into the grid [21]. Most existing harmonic standards are designed for devices operating at a fixed rated power [22]; however, smart charging introduces variability in charger operating points, fundamentally altering their harmonic behaviour and introducing new power quality concerns [23]. Although reducing charging rates allows more EVs to connect simultaneously, the accompanying harmonic emissions can accelerate transformer aging by increasing the hottest-spot temperature. The existing literature fails to account for harmonics impacts in transformer aging computations due to a lack of a dataset. The proposed framework bridges these domains by using real-world harmonic data and combining thermal modelling with THD-aware optimization, thereby offering a comprehensive assessment of EV smart charging impacts on transformer health and grid compliance. This way, DNOs will have better modelling and understanding of transformer aging during peak hours in the presence of smart EV charging. Therefore, system operators can make better decisions in managing existing assets and better plan for future expansions.

This study identifies two critical research gaps. First, while transformer aging under EV charging is well documented (e.g., [24], [25]), the specific impact of harmonics introduced by variable smart charging rates is overlooked, largely due to the lack of real-world harmonic datasets. Yet, as low-rate charging becomes prevalent in domestic settings, understanding these harmonic effects is essential. Here, this study utilizes real-world harmonic data to assess transformer aging and propose a rule-based management strategy to extend asset life.

Second, maintaining THD within regulatory thresholds in EV charging stations remains a complex challenge. Standards such as IEEE 519-2022 impose strict THD limits (e.g., 5%), yet smart charging implementations often ignore harmonic emissions, focusing solely on load balancing [26]. The trade-off between low charging power (favorable for grid loading) and high harmonic content has not been systematically addressed. This paper fills this gap by introducing a harmonics-aware smart charging framework that enforces THD limits during optimization.

The scientific aim of this work is to assess the impact of harmonics from smart EV charging on transformer aging and power quality. The subject of the research is the development of rule-based and optimization-based harmonics-aware charging strategies that integrate real-world harmonic measurements into both thermal models and charging algorithms. This study extends the existing research by combining transformer lifetime assessment with harmonics-aware optimization, providing a novel framework that bridges asset health, power quality compliance, and practical EV charging management. In parallel, arising studies on EV harmonic aggregation and stochastic fleet behaviour highlight that phase-angle diversity and heterogeneous charger technologies can lead to partial cancellation or reinforcement of specific harmonic orders [27], [28], [29]. Similarly, recent transformer thermal modelling efforts incorporate frequency-dependent loss factors under distorted load currents to improve aging predictions [30], [31]. These works collectively reinforce the importance of harmonics-aware smart charging, motivating this study's integrated thermal and THD-constrained optimization framework.

B. CONTRIBUTIONS

This article brings a new look to the existing literature in the following areas: (i) integrating real-world EV harmonic data into transformer aging assessments through a Monte Carlo-based framework, (ii) proposing a rule-based harmonics-averse charging management strategy for distribution transformers, (iii) introducing a harmonics-aware optimization approach that incorporates polynomial-based THD models and discrete current constraints via Water Filling algorithm, and (iv) demonstrating through case studies how these methods enable cost-effective compliance with harmonic standards without requiring expensive technological solutions such as active filters.

This paper makes the following key contributions:

- A Monte Carlo-based probabilistic assessment of transformer aging under smart EV charging using real-world harmonic data is conducted in accordance with IEEE C57.91 and IEEE C57.110 guidelines. Parameters such as hottest-spot temperature, aging acceleration factor, and estimated lifetime are evaluated for a 160 kVA transformer under varying charging rates.
- A rule-based harmonics-averse EV charging management strategy is proposed to guide system operators in prioritizing charging rates that balance grid needs and transformer health during peak hours.
- A harmonics-aware smart charging optimization framework is developed for commercial charging facilities, ensuring THD remains below prescribed limits (e.g., 5%). Regression models of THD as a function of charging rates are used for tractable closed-form representations within optimization.
- A Water Filling algorithm is applied to address discontinuities in charging rates (e.g., infeasibility between 0

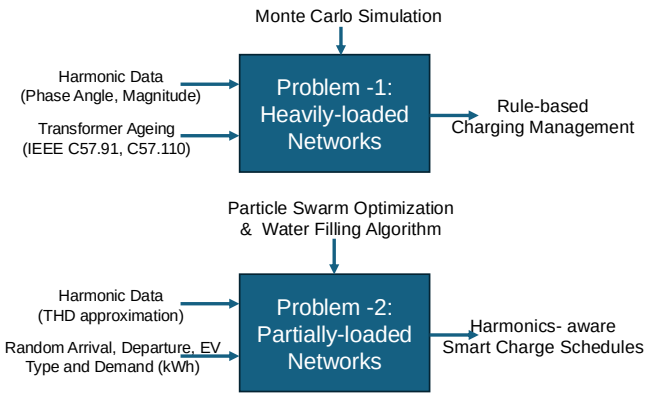


FIGURE 1: Overview of the two research problems addressed in this paper.

and 1.38 kW per phase), and post-processing methods are developed to enforce practical charging constraints.

- Detailed case studies demonstrate how THD compliance is achieved by exploiting EV time flexibility, either by deferring specific EV charging or prioritizing higher charging rates associated with lower harmonic emissions, thereby avoiding costly technological solutions such as active filters.

Figure 1 summarizes the two research problems. The first (Section IV) leverages EV harmonic data and IEEE guidelines within a Monte Carlo framework to evaluate transformer aging under various smart charging configurations. This analysis offers network operators insights into which charging rates to prioritize. The second problem (Section V) formulates a harmonics-aware optimization problem solved via particle swarm optimization, using the arithmetic mean of individual EV THDs as a conservative upper bound approximation. This enables flexible and practical smart charging without relying on discrete experimental rates alone. Both problems provide cost-effective, practical solutions to harmonics challenges in EV integration.

The remainder of this paper is organized as follows: Section II reviews the relevant literature on smart charging and harmonic impacts. Section III describes the data collection methodology and experimental setup. Section IV presents the transformer loss and aging model. Section V formulates the harmonics-aware optimization problem and outlines the solution strategy. Section VI evaluates the performance of the proposed framework under realistic scenarios, and Section VII concludes the paper with key findings and future directions.

II. LITERATURE REVIEW

The literature relevant to this work can be broadly categorized into two streams: (i) the impact of harmonics on distribution networks and transformer aging, and (ii) smart charging strategies for EVs.

At the distribution level, a significant body of research has focused on minimizing peak demand and maintaining voltage within operational limits by allocating reduced charging power to EVs connected to the same network [32]. Such studies frequently lead to assessments of EV hosting capacity, which indicates the maximum number of EVs that can be charged concurrently without necessitating costly network reinforcements [33]. However, with increasing EV penetration, existing infrastructure will inevitably require transformer and feeder upgrades. For instance, a recent study for California projects that by 2035, 50% of feeders will require upgrades, a figure expected to rise to 67% by 2045 [34]. Additionally, California's distribution network will need a capacity increase of 25 GW by 2045, demanding an investment ranging from $6 to $20 billion.

As most EV charging occurs within low-voltage residential and on-street networks, where typical AC chargers demand more power than an average household, distribution transformers are increasingly subjected to overloading risks [35]. This is particularly critical as most EV drivers arrive home and begin charging during peak residential demand periods, exacerbating grid congestion. Although transformer aging under EV charging has been extensively studied (e.g., [24], [25]), these works largely ignore the role of charging-level-dependent harmonic emissions (due to lack of datasets). Moreover, smart charging is often simplified to scheduling start and stop times without accounting for the dynamic harmonic behavior associated with varying charging rates.

For instance, [24] conducts a probabilistic assessment of transformer aging due to EV charging using IEEE C57.91-1995 guidelines. A 25 kVA pole-mounted distribution transformer is analyzed under uncontrolled peak-time EV charging scenarios in two cities, Vermont and Phoenix, to assess the combined effects of EV loading and ambient temperature. Results show that uncontrolled charging can lead to transformer loading as high as 40 kVA (60% above the rated capacity), significantly shortening transformer life. Similarly, [25] examines the impacts of residential EV charging on 25 kVA, 37.5 kVA, and 75 kVA transformers in Toronto, Canada, using real-world travel data to generate charge demand profiles. A scheduling algorithm is proposed to reduce transformer aging by limiting overloading, yet neither study considers the harmonic distortions introduced by variable EV charging rates. In contrast, the present study adopts a more comprehensive approach where smart charging strategies incorporate varying charging rates and their corresponding harmonic impacts.

Distribution transformers play a critical role in delivering electricity at appropriate voltage levels to residential, commercial, and industrial consumers. With the accelerating pace of EV adoption, many studies have highlighted the adverse effects of sustained or frequent transformer overloading on asset longevity [36], [37]. Parallel research has demonstrated that harmonic emissions exacerbate thermal stress and accelerate insulation degradation, thus compounding transformer aging [38], [39]. More recent works extend this perspective by quantifying frequency-dependent winding and stray losses under EV-driven harmonic spectra, demonstrating accelerated thermal aging even when RMS currents remain

within nameplate limits [40], [41], [42]. Parallel research also explores harmonic diversity across multiple EVs, reporting aggregation effects that depend on phase imbalance, charger type, and charging power levels [17], [43].

Recent studies further emphasize that harmonics generated by EV charging are emerging as a critical bottleneck in determining EV hosting capacity in distribution networks. For example, [44] demonstrates that depending on circuit topology, harmonic emissions may become a limiting factor even before thermal transformer limits are reached. Similarly, [45] presents a methodology to calculate EV hosting capacity in the presence of ultrafast chargers in Dutch distribution networks, revealing harmonics as a significant constraint on network expansion.

Smart charging is increasingly integrated into operational practices at AC charging stations due to its multifaceted benefits for both grid operators and EV owners. It can support the energy transition by shifting EV demand to periods of high renewable generation [46], [47], provide balancing reserves for transmission system operators [48], and help consumers reduce charging costs via time-of-use tariffs [49]. Beyond these operational benefits, recent research also highlights the role of smart charging in broader urban energy strategies. For instance, Lewicki et al. [50] propose integrating smart charging into the smart city concept, emphasizing the synergy between electromobility and renewable energy sources, where both the demand and supply sides of the future energy transformation are recipients of the benefits.

Smart charging has also gained attention as a key enabler in planning charging infrastructure and ensuring grid-aware operation. For instance, [51] proposes a joint optimization model for shared EVs that combines infrastructure siting with smart charging strategies such as time-of-use pricing and vehicle-to-grid (V2G) support. Their results indicate that incorporating V2G can reduce charging costs by up to 35%, with typical discharging contributing approximately 42% of the daily charging demand. Similarly, [52] develops a multi-objective charge scheduling framework for EV charging stations integrated with photovoltaic (PV) generation and battery energy storage systems. Their approach improves cost efficiency while mitigating peak load stress on the distribution network, highlighting the value of coordinated smart charging in renewable-rich, distributed energy environments.

Fast charging, associated with higher power levels, is favoured for user convenience but can place considerable stress on local distribution grids due to sharp increases in peak demand, often requiring costly infrastructure upgrades [53]. In contrast, slow and moderate AC charging is better aligned with smart charging strategies, as its lower power levels enable flexible scheduling and easier integration with grid limits and renewable sources [54].

Harmonic emission problems are typically associated with AC charging (generally up to 22 kW), as most AC chargers are located in public or private settings where grid capacity is limited and vehicles tend to remain connected for extended periods [24]. In such cases, smart charging strategies can effectively leverage EV flexibility to manage power quality [25]. In contrast, implementing smart charging, particularly through charging rate modulation, is generally not practical at rapid DC charging stations. These stations are designed to deliver energy as quickly as possible, and in many cases, drivers are charged parking fees during the charging session, further discouraging any delay [44]. Charge rate modulation is occasionally employed for demand management, such as during peak hours at Tesla Superchargers. In these instances, a central controller dynamically allocates power modules (for example, 25 kW each) among EVs to balance the load when multiple vehicles are charging simultaneously. However, since each module operates at its rated power, harmonic distortion is not a significant concern in such systems.

Ref. [26] illustrates a real-world smart charging implementation at a university campus equipped with various charger types. Here, smart charging dynamically allocates charging schedules based on vehicle arrival, departure, and energy demand, effectively leveraging user flexibility. However, a key practical issue arises: in AC chargers, charging rates below 6 A (1.38 kW in EU/UK) are typically infeasible, creating a discontinuity in possible charging rates. In [26], this issue is addressed by allocating a minimum of 6 A to each connected EV, prioritized by user flexibility and deadline constraints until infrastructure capacity is reached. Nonetheless, this approach leaves unresolved the broader question of how to systematically and optimally distribute charging rates under such constraints. This paper addresses the discontinuity challenge through the application of a Water Filling Algorithm, which equitably distributes available capacity while respecting all charging constraints, including infeasible regions. Notably, most existing optimization models overlook this discontinuity when designing smart charging strategies [55].

Another key consideration in smart charging is user flexibility, as highlighted in [56], [57], [58], where overstay periods, instances when an EV remains connected but is not actively charging, are shown to represent a significant share of total connection time. Exploiting this latent flexibility can enable additional control levers to mitigate harmonic emissions, for instance, by selectively pausing charging sessions for certain vehicles to stay within THD limits.

Finally, this study builds upon prior work (e.g., [21], [59]), which analyzed harmonic emissions from eight mainstream EV models and found that most vehicles exceed individual harmonic amplitude and THD limits set by standards such as IEC 61000-3-X and IEEE 519-2014. The data and insights from these studies form the empirical foundation for the advanced modelling and optimization frameworks proposed in this paper. The following section provides a detailed overview of the harmonic data analysis, which underpins the methodologies presented in subsequent sections.

III. SMART CHARGING HARMONIC EMISSIONS

The input data used in this study were obtained from a series of controlled smart charging experiments, with detailed methodologies and preliminary analyses presented in prior

work [21]. A total of eight pure EV models were analyzed to provide a comprehensive representation of diverse manufacturers, battery sizes, and on-board charger technologies. The tested vehicles included: Nissan Leaf e+, Peugeot e-208, Peugeot e-2008, Renault Zoe R90, Renault Zoe ZE50, Tesla Model Y Long Range, VW ID.3 Pro, and VW ID.4 Pro. It is important to note that this paper deals with current harmonics, as voltage harmonics are fairly consistent throughout different charging rates as given in [21].

Among these vehicles, seven of them support three-phase charging, whereas only the Nissan Leaf e+ supports single-phase charging. This distinction is important because charger topology directly shapes the harmonic spectrum. Single-phase chargers typically employ a 2-pulse diode bridge rectifier, which generates harmonics of the form $h = 2n \pm 1$, leading to strong low-order components such as the 3rd, 5th, and 7th. In contrast, three-phase chargers generally use 6-pulse rectifiers, which produce harmonics of the form $h = 6n \pm 1$, with the 5th, 7th, 11th, and 13th orders being dominant. Moreover, in balanced three-phase systems, triplen harmonics (multiples of 3, e.g., 3rd, 9th, 15th) cancel out in the line currents, which explains their absence in the measured spectra. Consistent with this theory, the measurements showed that the single-phase Nissan Leaf e+ exhibited stronger low-order harmonics (particularly the 3rd order), whereas the three-phase EVs followed the expected 6-pulse pattern with 5th, 7th, 11th, and 13th harmonics dominating. By incorporating both single- and three-phase chargers, the dataset strengthens the robustness and realism of the harmonic models developed in this study.

In addition to these observations, it is important to note that three-phase on-board chargers typically employ two-level full-bridge power-factor-correction (PFC) topologies that avoid twice-line-frequency power pulsation. When paired with appropriately designed L or LCL filters, these converters can satisfy harmonic limits efficiently and without inducing phase imbalance [60]. Conversely, single-phase chargers inherently experience pulsating power at twice the grid frequency, requiring large DC-link capacitors to stabilize voltage. If not adequately controlled, zero-crossing current spikes may occur, increasing THD. The dominant topology for single-phase automotive PFC stages is the totem-pole bridgeless configuration, which offers high efficiency but remains more susceptible to harmonic distortion and phase imbalance, especially under light-load or grid-distorted conditions [61].

Additionally, switching frequency and modulation strategy influence harmonic distribution: lower-frequency PWM tends to concentrate energy in lower-order harmonics, while higher-frequency PWM spreads harmonic components into higher orders, shifting mitigation considerations toward electromagnetic compatibility [62]. Therefore, incorporating both single-phase and three-phase charging architectures in our dataset ensures that the harmonic models reflect realistic residential and commercial charging environments, enhancing the robustness and accuracy of transformer aging and power-quality assessments.

All vehicles were charged via a Fronius Wattpilot AC smart charger, which was controlled using its proprietary mobile platform [63]. The charging current was varied incrementally in 1 A steps, covering each EV's minimum (e.g. 6 A) to maximum allowable charging current. This experimental design enabled a granular characterization of EV charging behavior across a broad range of real-world operating conditions. Note that in the next sections, charging current and power are used interchangeably and per phase power (in kW) can simply be computed by multiplying the charging current with the voltage level, i.e. 230 V. For instance, the minimum allowable charging power becomes 6 A $\times$ 230 V = 1.38 kW.

To accurately capture electrical parameters, a Yokogawa WT500 precision power analyzer [63] was employed, recording data at a 1-second resolution. Measured parameters included root mean square (RMS) current and voltage, fundamental frequency components, and detailed harmonic spectra, including both amplitude and phase of individual harmonic orders. This high-resolution dataset provided a robust foundation for evaluating the influence of variable charging rates on harmonic emissions and transformer loading.

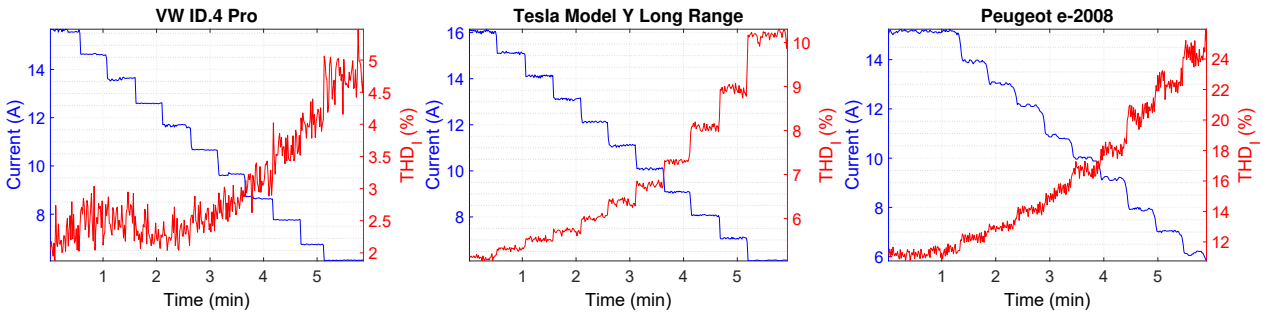
Figure 2 illustrates the current THD as a function of smart charging current for each tested EV. An inverse correlation between charging current and harmonic emissions is observed since higher charging rates generally result in lower THD levels. Although the Tesla Model Y supports high charging rates, low-rate scenarios (6-10 A) are used to reflect typical residential operation, where chargers often run at reduced currents. These low rates produce higher harmonic distortion, making them critical for realistic aging analysis. To quantify this trend, a statistical analysis was conducted, converting charging current (A) to per-phase charging power (kW) for ease of representation and analysis. The maximum single-phase charging power is calculated as $3.68\,\text{kW} = 230\,\text{V} \times 16\,\text{A}$, while three-phase chargers reach up to 11 kW. A minimum charging current of 6 A (1.38 kW) is maintained across all tested vehicles as stated earlier.

To model the harmonic behaviour as a function of charging power, a quadratic polynomial fitting was applied to each vehicle dataset

$$THD(x) = p_1 x^2 + p_2 x + p_3, \tag{1}$$

where x denotes the per-phase charging power in kW, and p_1, p_2, and p_3 are the polynomial coefficients. This closed-form expression enables the integration of THD modelling into optimization frameworks, facilitating advanced smart charging algorithms while capturing the essential nonlinear relationship between power and harmonic emissions.

As shown in Table 1, the fitted polynomial models exhibit high accuracy, with coefficient of determination (R^2) values consistently above 0.86, and in several cases exceeding 0.99, demonstrating a strong fit. These convex second-order polynomials serve as foundational elements in subsequent optimization formulations (Section V). Importantly, this approach overcomes limitations associated with raw experimen-



(a) VW ID.4 Pro (b) Tesla Model Y Long Range (c) Peugeot e-2008

FIGURE 2: THD versus Smart Charging Current for Different Electric Vehicles

TABLE 1: Quadratic polynomial fitting parameters for total harmonic distortion (THD) as a function of charging power.

EV Model	p_1	p_2	p_3	R^2
Renault Zoe R90	0.26	-3.41	16.14	0.9675
Peugeot e-208	0.82	-5.39	11.42	0.8634
Nissan Leaf e+	0.41	-4.16	13.98	0.9235
VW ID.3 Pro	0.73	-4.63	9.36	0.9734
Renault Zoe ZE50	0.48	-5.26	18.92	0.9681
VW ID.4 Pro	0.86	-5.26	10.28	0.9167
Tesla Model Y Long Range	1.00	-7.17	18.07	0.9920
Peugeot e-2008	1.94	-15.76	42.74	0.9928

tal datasets, which restrict charging rates to discrete values due to equipment resolution. By adopting a continuous polynomial model, a broader and more flexible set of charging rates can be evaluated in optimization, better reflecting real-world deployment scenarios.

Figure 3 presents the overall correlation between per-phase charging power and THD, demonstrating the general downward trend of THD as charging power increases. This relationship forms the core of the harmonics-aware smart charging framework proposed in this study, enabling EV charging schedules that respect both grid capacity and harmonic distortion limits.

IV. TRANSFORMER AGING IN HEAVILY LOADED NETWORKS

As discussed in the first section, harmonic emissions from smart EV charging introduce additional stress on heavily loaded distribution networks, particularly on transformers operating near their rated capacity. In addition to electrical loading, thermal effects, driven by internal losses, play a critical role in assessing transformer aging.

The total transformer losses, P_{TL}, comprise no-load losses (P_{NL}) and load-dependent losses (P_{LL}), which include copper losses (P_{CL}) and stray losses (P_{SL}) [64], [65]:

$$P_{TL} = P_{NL} + P_{LL} = P_{NL} + (P_{CL} + P_{SL}). \tag{2}$$

Stray losses can be further decomposed into winding eddy-current losses (P_{ECL}) and other stray losses (P_{OSL}):

$$P_{LL} = P_{CL} + P_{SL} = P_{CL} + (P_{ECL} + P_{OSL}). \tag{3}$$

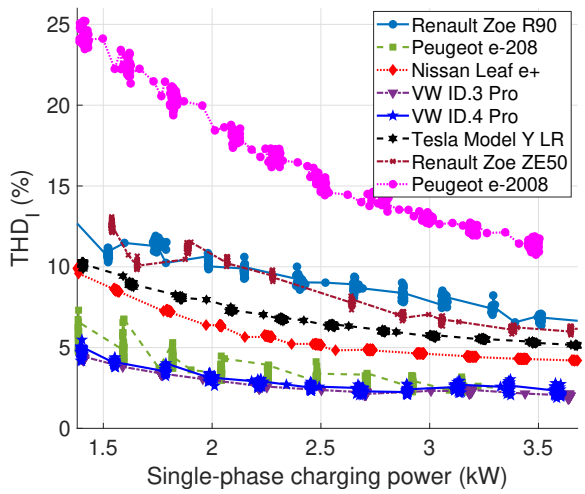


FIGURE 3: Relationship between EV charging power and THD (per phase).

These losses contribute to increased top-oil and hot-spot temperatures, accelerating insulation degradation and shortening transformer life.

A fundamental consequence of harmonic distortion is the rise in the RMS value of the load current, which increases thermal loading. The RMS current under harmonic conditions is calculated as,

$$I_L^h = \sqrt{\sum_{h=1}^{h_{\max}} I_h^2}, \tag{4}$$

where I_h denotes the RMS current component of the h-th harmonic, and $h_{\max}$ is the highest harmonic order considered. Even low-order harmonics can significantly increase I_L^h, leading to accelerated transformer aging due to thermal stress.

The components of load loss, P_{LL}, increase with the square of the current, but their dependence on frequency varies. Copper losses (P_{CL}) slightly increase at higher frequencies due to the skin effect, though this is minor for small conductors. Eddy-current losses in windings (P_{ECL}) grow roughly with the square of frequency, while those in structural parts (P_{OSL}) follow a sub-quadratic trend, typically with a fre-

quency exponent of 0.8. These frequency-dependent effects, driven by stray magnetic fields, have a significant impact on thermal stress. Accounting for these frequency-dependent effects, harmonic-distorted load losses in a transformer can be mathematically expressed as:

$$P_{LL}^{h} = P_{CL}^{h} + P_{ECL}^{h} + P_{OSL}^{h}, \tag{5}$$

which can equivalently be written as:

$$P_{LL}^{h} = (P_{CL}^{r} + F_{HL}^{ec} P_{ECL}^{r} + F_{HL}^{os} P_{OSL}^{r}) \times I_{L,pu}^{2}. \tag{6}$$

In these equations, P_{CL}^{r} represents the reference copper losses at a nominal frequency, P_{ECL}^{r} and P_{OSL}^{r} denote the reference eddy-current and other stray losses, respectively, while F_{HL}^{ec} and F_{HL}^{os} are harmonic loss factors that quantify the impact of harmonic distortion on P_{ECL} and P_{OSL}. These baseline losses are provided by the transformer manufacturer for rated conditions and are used to scale loss increases under harmonic loading. The term $I_{L,pu}$ is the per-unit load current, normalizing the load current relative to the transformer's rated current.

In the context of harmonic distortion analysis, the superscript h denotes the presence of harmonics, whereas the superscript r refers to the rated conditions associated with loading and frequency. To quantify the impact of harmonics on transformer performance, the per-unit harmonic load current, $I_{L,pu}$, is defined as the ratio of the harmonic load current to the rated load current. This relationship is mathematically expressed as:

$$I_{L,\mathrm{pu}} = \frac{I_L^h}{I_L^r} \tag{7}$$

where I_L^h represents the RMS harmonic load current, and I_L^r denotes the RMS load current under rated conditions. The per-unit system normalizes the harmonic load current, facilitating comparing and analysing harmonic effects relative to the transformer's rated performance.

Furthermore, the harmonic loss factors for winding eddy currents and other stray losses are denoted by F_{HL}^{ec} and F_{HL}^{os}, respectively. These factors quantify how harmonics amplify eddy-current losses in the windings (F_{HL}^{ec}) and stray losses in structural parts (F_{HL}^{os}). The harmonic loss factors are calculated using the following equations:

$$F_{HL}^{ec} = \frac{\sum_{h=1}^{h_{\max}} I_h^2 \times h^2}{\sum_{h=1}^{h_{\max}} I_h^2}, \tag{8}$$

$$F_{HL}^{os} = \frac{\sum_{h=1}^{h_{\max}} I_h^2 \times h^{0.8}}{\sum_{h=1}^{h_{\max}} I_h^2}, \tag{9}$$

where $I_h$is the RMS load current associated with the h-th harmonic order, and $h_{\max}$ is the highest harmonic order considered in the analysis.

The harmonic loss factor for winding eddy currents, F_{HL}^{ec}, is approximately proportional to the square of the harmonic order. This indicates that higher-order harmonics have a more significant impact on eddy-current losses within the transformer windings. Conversely, the harmonic loss factor for other stray losses, F_{HL}^{os}, exhibits a sub-quadratic dependence on the harmonic order, characterized by an exponent factor of 0.8. This suggests that while higher-order harmonics still contribute to stray losses, their impact is less pronounced compared to eddy-current losses. In practice, however, lower-order harmonics (e.g., 3rd, 5th, and 7th) tend to dominate the overall loss increase because of their relatively larger amplitudes, which significantly elevate the RMS current and heating within the transformer. Therefore, both low-order harmonics, due to magnitude, and higher-order harmonics, due to frequency weighting in F_{HL}^{ec} and F_{HL}^{os}, play a role in accelerating transformer aging.

A. TRANSFORMER TEMPERATURE RISE MODEL

The top-oil temperature (θ_{TO}) of the transformer can be determined by first estimating the load losses under harmonic currents and then applying the following relationship:

$$\theta_{\mathrm{TO}} = \theta_A + \Delta\theta_{\mathrm{TO}}, \tag{10}$$

where $\Delta\theta_{TO}$ represents the rise in top-oil temperature above the ambient temperature (θ_A). The temperature rise, $\Delta\theta_{TO}$, is calculated as:

$$\Delta\theta_{\mathrm{TO}} = \Delta\theta_{\mathrm{TO}}^{r} \times \left(\frac{P_{LL}^{h} + P_{NL}}{P_{LL}^{r} + P_{NL}}\right)^{0.8}, \tag{11}$$

where $\Delta\theta_{TO}^{r}$ is the temperature gradient between the top-oil layer and the ambient temperature under rated conditions, as specified by the transformer manufacturer, in this context, P_{LL}^{h} denotes the load losses under harmonic currents, P_{LL}^{r} represents the rated load loss, and P_{NL} stands for the no-load loss. As harmonic currents increase, P_{LL}^{h} contributes to higher overall load losses, thereby elevating $\Delta\theta_{TO}$ and subsequently θ_{TO}. Elevated top-oil temperatures can accelerate the aging of transformer insulation, reduce the dielectric strength of insulating materials, and ultimately diminish the operational lifespan of the transformer.

Having determined the top-oil temperature (θ_{TO}), the hot-spot temperature (θ_{HS}) within the transformer winding can be evaluated using the following relationship:

$$\theta_{\mathrm{HS}} = \theta_{\mathrm{TO}} + \Delta\theta_{\mathrm{HS}}, \tag{12}$$

where $\Delta\theta_{HS}$ represents the temperature rise of the hot spot above the top-oil temperature. The temperature rise in the hot spot, $\Delta\theta_{HS}$, is calculated as:

$$\Delta\theta_{\mathrm{HS}} = \Delta\theta_{\mathrm{HS}}^{r} \times \left(\frac{P_{LL}^{h}}{P_{LL}^{r}}\right)^{0.8}, \tag{13}$$

where $\Delta\theta_{HS}^{r}$ denotes the temperature difference between the hot spot and the top-oil layer under rated loading conditions. Elevated hot-spot temperatures are particularly detrimental as they directly affect the aging rate of the transformer's insulation materials. Higher θ_{HS} accelerates the thermal degradation processes, reducing the dielectric strength and overall lifespan of the transformer [64], [38]. Additionally, excessive θ_{HS}

can lead to uneven heating within the windings, promoting the formation of localized hot spots that further compromise the integrity of the transformer's structural and electrical components [42].

B. TRANSFORMER LIFETIME MODELLING

For a transformer rated with a 55 °C average winding temperature rise, the aging acceleration factor, F_{AA}, is defined to be 1.0 at a reference hot-spot temperature (HST) of 110 °C. This factor quantifies the rate at which transformer insulation ages relative to its normal operating conditions. Mathematically, F_{AA} is directly proportional to the HST and exceeds 1.0 when the HST surpasses 110 °C, thereby indicating an accelerated rate of insulation degradation. The relationship governing F_{AA} is expressed as:

$$F_{AA} = \exp\left(\frac{15000}{383} - \frac{15000}{\theta_{\text{HS}} + 273}\right), \tag{14}$$

where θ_{HS} is the hot-spot temperature.

Under rated conditions, where the HST is maintained at 110 °C, the aging acceleration factor F_{AA} equals 1.0, signifying that the transformer insulation ages at its standard rate. The normal insulation life, Life_{norm}, for a system operating under these conditions is established to be 20.55 years (equivalent to 180,000 hours) [64]. This baseline serves as a reference point for assessing the real-life expectancy of transformers subjected to varying thermal stresses induced by harmonic distortions.

Using the aging acceleration factor, the real-life expectancy of a distribution transformer, Life_{real}, can be determined by adjusting the normal insulation life based on the actual operating conditions. This relationship is given by:

$$\text{Life}_{\text{real}} = \frac{\text{Life}_{\text{norm}}}{F_{AA}} \tag{15}$$

Here, Life_{real} represents the estimated transformer lifespan (in years), accounting for thermal stress caused by harmonic-induced losses. As F_{AA} increases above 1.0, Life_{real} decreases accordingly, indicating accelerated transformer aging.

F_{AA} quantifies how harmonic distortions, primarily from EV smart charging, impact transformer longevity. Elevated harmonic currents raise P^h_{LL} and the hot-spot temperature θ_{HS}, thereby shortening service life and increasing maintenance costs. To summarize the complete transformer aging assessment process, Figure 4 provides a flowchart outlining the sequential steps from harmonic and thermal inputs to transformer lifetime estimation. By incorporating F_{AA} into performance assessments, it becomes possible to predict long-term impacts and design smart charging strategies that limit θ_{HS}, ultimately mitigating transformer aging and supporting grid reliability amid rising EV adoption.

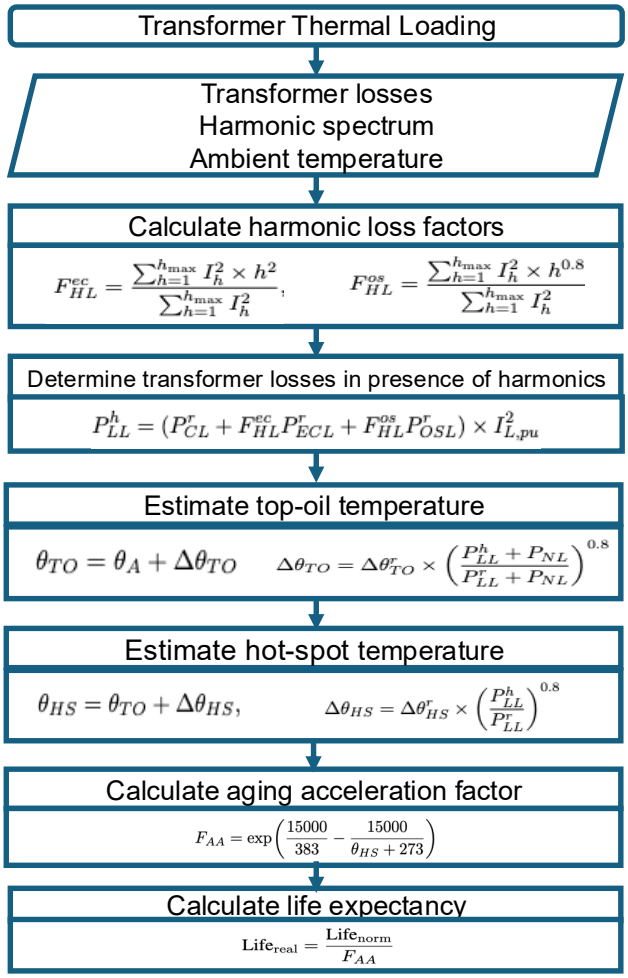


FIGURE 4: Flowchart of the transformer aging assessment process.

V. TOTAL HARMONIC DISTORTION IN PARTIALLY-LOADED NETWORKS

While the previous section focuses on the impacts of smart charging harmonics on transformer aging, harmonic emissions from smart EV charging introduce additional challenges in partially loaded networks. Specifically, EV station operators must ensure that THD levels remain within the limits imposed by relevant standards, such as IEEE 519-2014 and IEC 61000-3-X, which typically restrict THD to 5% (and to different values depending on circuit configuration). Given that smart charging-based scheduling is now a common feature of modern charging infrastructure, this section presents a comparative analysis of smart charging frameworks, with and without harmonic awareness, using a large-scale EV parking lot as a case study representative of partially loaded networks.

A critical component of the proposed harmonics-aware smart charging framework is the modelling and aggregation of harmonic content from multiple EVs, which is detailed in the following subsection.

A. HARMONICS MODELLING

While modelling the harmonic emissions of an individual EV is relatively straightforward using the polynomial fitting parameters provided in Table 1, estimating the total harmonic emissions from multiple EVs, whether identical

models charging at different rates or diverse models operating at arbitrary charging levels, introduces significant complexity. This complexity arises from the absence of a closed-form solution to account for both the amplitude and phase angle of individual harmonics.

The total harmonic emissions from multiple EVs must be calculated using vector algebra, where each harmonic order is treated as a phasor with both magnitude and phase. An individual harmonic component can be expressed as $A \cdot e^{j\theta}$, or equivalently in rectangular form as $A \cdot (\cos(\theta) + j\sin(\theta))$, where A is the magnitude and θ is the phase angle. When multiple harmonic sources are present, their components are summed (in vector form), and the resulting vector is used to determine THD and the aggregated form is computed as:

$$THD = \frac{\sqrt{\sum_{h=2}^{H} I_h^2}}{I_1} \times 100\%, \tag{16}$$

where h denotes the harmonic order, H denotes the highest harmonic order considered in the analysis (49th order in this study), I_h is the magnitude of the hth harmonic component, and I_1 is the magnitude of the fundamental frequency component.

However, integrating such vector summation directly into optimization frameworks presents significant challenges, as individual harmonic magnitudes I_h and phase angles cancel each other in a random fashion and the literature lacks models that capture harmonic cancellation stemming from multiple harmonic sources. Without closed-form expressions, an optimization-based smart charging framework would require an exhaustive search over all charging configurations, which is computationally prohibitive for real-time or large-scale applications.

To address this issue, a tractable approximation is proposed that provides a conservative upper bound for THD. As empirically demonstrated in [21], when multiple EVs charge simultaneously, a natural harmonic cancellation effect occurs, reducing the aggregated THD relative to the sum of individual emissions. Notably, the observed total THD remains consistently below the arithmetic mean of individual THD levels, due to phase angle diversity that causes partial harmonic cancellation. This is mathematically explained by the phasor summation process, where harmonic components are scaled by $\cos(\theta)$ and $\sin(\theta)$, resulting in reduced net amplitude.

Therefore, the arithmetic mean of approximation of individual THDs offers a computationally efficient and conservative approximation of total THD. This approximation allows the optimization to avoid expensive vector-based summations while ensuring that the THD constraint is respected. Mathematically, if THD_i denotes the THD contribution of the ith EV, the following relation holds:

$$THD_{\text{total}} \lessapprox \frac{1}{N} \sum_{i=1}^{N} THD_i, \tag{17}$$

where N is the total number of simultaneously charging EVs.

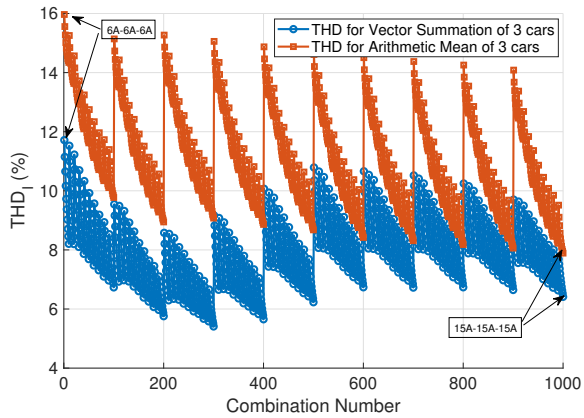


FIGURE 5: Vector summation versus arithmetic mean comparison of THD with 3 EVs charging

Figure 5 illustrates the comparison between the arithmetic mean and vector summation of harmonic emissions for three EV models (Tesla Model Y, Peugeot e-2008, and Renault Zoe R90), charging at identical rates. The x-axis represents charging combination indices, ranging from minimum (6A, 6A, 6A) to maximum (15A, 15A, 15A) currents for three EVs. The results confirm that the arithmetic mean consistently exceeds the vector summation, validating its use as a conservative upper bound.

This approximation greatly simplifies the optimization of harmonics-aware smart charging. Rather than explicitly computing vector-based harmonic summations for every combination of EV charging rates, the optimization problem can instead enforce THD constraints based on the arithmetic mean of the fitted polynomial models (as described in Section III). This enables practical deployment of harmonics-aware smart charging in real-world networks, including large parking lots and public charging stations, without incurring prohibitive computational costs. To that end, this section establishes a mathematically tractable approach to capture the harmonic emissions of multiple EVs under smart charging, forming the basis of the optimization framework presented in the subsequent section.

B. HARMONICS-AWARE SMART CHARGING

This section presents the mathematical formulation for two smart charging optimization problems. The first problem represents a classical smart charging framework with realistic charging rates, while the second problem extends this formulation to explicitly incorporate THD limits.

Let $i = \{1, 2, \ldots, N\}$ denote the index of EVs arriving at a charging station that operates over discrete time slots $t = \{1, 2, \ldots, T\}$. Each EV arrives at time $a_i \in T$ and departs at time $d_i \in T$, where the minimum charging duration is given by $d_i - a_i$, and must satisfy a strictly positive value M, $\forall i$, to ensure meaningful participation. Thus, the first constraint ensures a minimum connection time and is expressed as

$$d_i - a_i \geq M, \quad \forall i \tag{18}$$

Let $k = \{1, 2, \ldots, K\}$ denote the index of EV types (e.g., $k = 1$ for Tesla Model Y). The charging demand of each EV i, expressed in kWh, is denoted as D_i, and the decision variable for smart charging is represented by the charging vector $\boldsymbol{x_i} \in \mathbb{R}^{1\times T}$, $\forall i$. Additionally, let $\boldsymbol{x^t}$ denote the vector of charging rates across all EVs at time t.

Due to technical limitations, charging currents are constrained to values between 6 A (corresponding to $c_{min} = 1.38\,\text{kW}$ per phase) and 16 A (corresponding to $c_{max} = 3.68\,\text{kW}$ per phase), or to zero if not charging. Here, c_{min} and c_{max} reflect safety and hardware limits of the chargers, defining the operational charging window. Intermediate values between 0 A and 6 A are not feasible, reflecting a discontinuous feasible domain. Assuming all chargers are three-phase, but considering per-phase rates for analysis, the charging rate constraint is formulated as [66]:

$$x_i(t) \in \{0\} \cup [c_{\min}, c_{\max}], \quad \forall i, \forall t. \tag{19}$$

The charging power $x(t)$ optimized by the PSO algorithm corresponds to the per-phase charging rate (in kW) for each EV at each time step. These values are not computed analytically but are found iteratively by the PSO algorithm, which updates each candidate solution based on harmonic distortion, charger power limits, EV availability periods, and minimum demand satisfaction. The algorithm ensures convergence toward a solution that balances energy delivery objectives with power quality constraints.

Furthermore, the aggregated charging rates across all EVs at any time t are limited by a system-wide capacity constraint C, representing infrastructure constraints (e.g., transformer, cable limits). The capacity constraint is expressed as

$$\sum_{i=1}^{N} x_i(t) \leq C, \quad \forall t, \tag{20}$$

where C is typically set as a percentage of the total peak load capacity, e.g., $C = 0.8 \times N \times 3.68\,\text{kW}$, corresponding to a 20% reduction in infrastructure requirement due to smart charging coordination. This constraint highlights that without such coordination, network capacity would need to be significantly oversized to handle peak EV demand.

Additionally, to ensure service quality, a minimum energy delivery requirement $D_{\min}$ is imposed to guarantee that each EV receives at least a specified fraction of its requested demand D_i. This is written as

$$\sum_{t=a_i}^{d_i} x_i(t) \geq D_{\min} \cdot D_i, \quad \forall i. \tag{21}$$

A typical value for $D_{\min}$ is 0.9 (i.e., 90%), ensuring high customer satisfaction while allowing operational flexibility.

1) Optimization Problem I: Smart Charging without THD Constraints

The objective of the first optimization problem is to minimize the deviation between the requested demand D_i and the allocated charging schedule $\boldsymbol{x_i}$, ensuring fairness among EVs. Assuming equal weight for each EV, the objective function is defined as:

$$\sum_{i=1}^{N} \left(\frac{\sum_{t=a_i}^{d_i} x_i(t)}{D_i} - 1 \right)^2. \tag{22}$$

Thus, **Optimization Problem I** can be formulated as:

$$\begin{aligned} &\text{Minimize:} && (22) \\ &\text{Subject to:} && (18),\ (19),\ (20),\ (21). \end{aligned}$$

Note that since the optimization is performed over 1-hour intervals, the charging rate vector $\mathbf{x}(t)$ in kW is numerically equivalent to the energy delivered in kWh for each time slot. This allows direct comparison with the energy demand D in the constraint formulation.

2) Optimization Problem II: Smart Charging with THD constraints

The second optimization problem extends the first by including explicit constraints on THD. At each time slot t, the aggregated THD, computed as a function of the charging rates $\boldsymbol{x^t}$ and EV types $\boldsymbol{k^t}$ connected at time t, must not exceed a predefined limit h_{lim}, typically set to 5%. The THD constraint is expressed as:

$$\text{THD}(\boldsymbol{x^t}, \boldsymbol{k^t}) \leq h_{\text{lim}}, \quad \forall t. \tag{23}$$

In addition to minimizing demand deviation, the objective function includes a penalty for any violation of the THD constraint. The penalized objective function is given by:

$$\sum_{i=1}^{N} \left(\frac{\sum_{t=a_i}^{d_i} x_i(t)}{D_i} - 1 \right)^2 + \omega \sum_{t\in\bar{T}} \left(\text{THD}(x^t, k^t) - h_{\text{lim}}\right)^2, \tag{24}$$

where $\bar{T}$ is the set of time slots where the THD limit is violated, and ω is a large penalty coefficient ($\omega \gg 0$) used to ensure strong enforcement of the harmonic constraint.

Hence, **Optimization Problem II** is formulated as:

$$\begin{aligned} &\text{Minimize:} && (24) \\ &\text{Subject to:} && (18),\ (19),\ (20),\ (21),\ (23). \end{aligned}$$

In summary, while Optimization Problem I focuses on efficient and fair allocation of charging resources, Optimization Problem II advances this by embedding harmonic emission limits directly into the optimization, thereby enabling harmonics-aware smart charging in compliance with stringent power quality standards.

C. PARTICLE SWARM OPTIMIZATION

Before selecting Particle Swarm Optimization (PSO) as the core algorithm, several alternative optimization techniques were tested, including Genetic Algorithms (GA) and gradient-based non-linear optimisation methods. However,

Algorithm 1 PSO-based Harmonics-Aware Smart Charging Optimization

Input: EV data (N, $\{a_i, d_i, D_i\}$, EV types), charging constraints ((19), (20), (21), (23)), PSO parameters (swarm size, max iterations, inertia w, acceleration coefficients c_1, c_2)
Generate EV arrival (a_i), departure (d_i), and demand (D_i) profiles
for each scenario (Optimization Problem I: $\omega = 0$; Optimization Problem II: $\omega > 0$) **do**
    Initialize swarm with random charging schedules x_p and velocities v_p
    Apply RepairSolution to each x_p to ensure feasibility
    Evaluate objective ((22) or (24)), set particle best positions p_{best}
    Identify global best position g_{best} among all p_{best}
    for $iter = 1$ to max iterations **do**
        for each particle p **do**
            Apply RepairSolution to x_p
            Evaluate objective, update p_{best} and g_{best} if improved
            Update velocity:

$$v_p \leftarrow wv_p + c_1(p_{\text{best}} - x_p) + c_2(g_{\text{best}} - x_p)$$

            Update position:

$$x_p \leftarrow x_p + v_p$$

            Enforce all constraints ((19), (20), (21), (23))
        end for
    end for
    Output: Optimal charging schedule $x_{\text{opt}} = \text{RepairSolution}(g_{\text{best}})$
end for

these approaches either failed to converge or did not yield feasible solutions, primarily due to the nonconvex, discontinuous nature of the feasible solution space imposed by THD constraints and discrete EV charging decisions. In contrast, PSO consistently produced feasible results and offered a robust balance between convergence reliability and computational efficiency, making it suitable for the harmonics-aware smart charging problem considered in this study.

To solve both optimization problems described above, PSO [67] method is chosen primarily to handle discontinuity in charging rates given in (19). The PSO algorithm is a search-based stochastic algorithm which explores the best tuning parameters to minimize an objective function. Firstly, P particles are distributed randomly in the solution space. At each iteration, each particle p moves to different location with a velocity and the best velocity direction (V_p) is recorded. At each iteration, velocity and direction information is shared with other particles and the maximum/minimum objective values recorded. Therefore, the next strategy (moving direction) is determined by weighted sum of cognitive (c_1) and social (c_2) coefficients. A detailed implementation of PSO for EV charging can be found in [68].

D. WATER FILLING ALGORITHM

Algorithm 1 presents the details of the PSO-based optimization framework used in this study. Within this algorithm, a critical subroutine, referred to as RepairSolution, is invoked to enforce all optimization constraints, including those outlined in (18), (19), (20), (21), and (23). Additionally, RepairSolution dynamically monitors actively charging EVs and updates THD contribution by removing EVs that are either temporarily turned off or have already departed.

To ensure that both the charging rate constraints (19) and minimum demand constraints (21) are satisfied, a "Water Filling" procedure is embedded within RepairSolution. This procedure operates as follows:

1) First, all charging rates below the minimum charging threshold $c_{\min}$ are set to zero:

$$x_i(t) = 0, \quad \text{if } x_i(t) < c_{\min}, \quad \forall i, t. \tag{25}$$

Charging power is set to zero if it falls below the minimum feasible threshold $c_{\min}$ = 1.38 kW.

2) If the cumulative charge does not meet the minimum demand $D_{\min} \times D_i$, the algorithm activates zero-charging slots by assigning them the minimum allowed rate $c_{\min}$, until either the demand is satisfied or all slots are active.

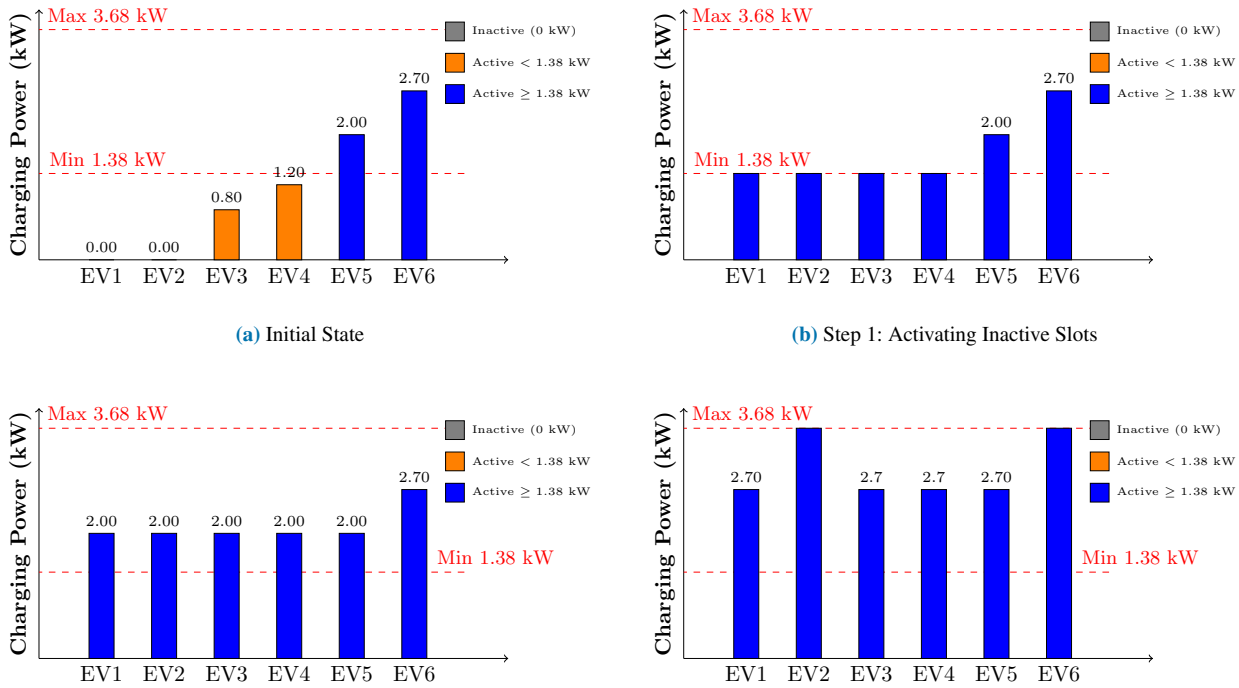
3) If the demand constraint remains unsatisfied, the algorithm incrementally increases all active slots, initially set at $c_{\min}$, in a uniform manner, up to $c_{\max}$, while ensuring that the aggregate charging rates at any time t do not exceed the system capacity constraint (20).

4) Finally, for each time slot t, the total THD is checked to ensure compliance with (23). If the THD exceeds the allowed limit h_{lim}, charging rates are selectively reduced, starting from the highest contributors, until the limit is respected.

This structured and iterative Water Filling process guarantees that charging solutions comply with critical constraints while prioritizing user demand satisfaction and harmonic limits.

Figure 6 illustrates an example application of the Water Filling Algorithm. In the initial state (Figure 6a), EV1 and EV2 are inactive (0 kW), while others are active at varying levels. Charging limits are set between $c_{\min} = 1.38$ kW and $c_{\max} = 3.68$ kW as per (19). Slots below $c_{\min}$ (e.g., EV3 at 0.8 kW, EV4 at 1.2 kW) are marked in orange, and those above are in blue (e.g., EV5 at 2.0 kW, EV6 at 2.7 kW).

In Step 1 (Figure 6b), all inactive and sub-threshold EVs are brought up to $c_{\min}$. Thus, EV1 and EV2 are activated at 1.38 kW, and EV3 and EV4 are raised to this minimum threshold. EV5 and EV6 remain unchanged, as they already exceed the minimum. In Step 2 (Figure 6c), if the minimum demand is still unmet, the algorithm uniformly increments all active charging slots until either the demand is satisfied or the maximum charging rate $c_{\max}$ is reached. For example, EV1 to EV5 are raised to 2.0 kW, and EV6 remains at 2.7 kW. In Step 3 (Figure 6d), the final charging distribution is shown where all EVs receive at least $c_{\min}$, some reaching $c_{\max}$. In this case, EV2 and EV6 are capped at 3.68 kW, and others receive approximately 2.7 kW, ensuring optimal use of available capacity while satisfying user demands and harmonic constraints.



(a) Initial State

(b) Step 1: Activating Inactive Slots

(c) Step 2: Incrementing Active Slots

(d) Step 3: Final Charging State

FIGURE 6: Step-by-step illustration of Water Filling Algorithm.

VI. CASE STUDIES

A. CASE I: HEAVILY LOADED NETWORKS

This section presents a case study in which a distribution transformer serves a congested urban area operating close to its rated capacity. The electrical loading of a distribution transformer is a key determinant in assessing its ability to accommodate simultaneous EV charging. The maximum number of EVs that can be charged concurrently, without exceeding the transformer's rated capacity, depends critically on the selected smart charging rate. At lower charging currents, a greater number of EVs can be supported simultaneously, a favorable scenario for system operators. However, as shown in previous sections, lower charging rates introduce higher harmonic emissions, intensifying thermal stress on the transformer. This case study analyzes the trade-off between the number of EVs that can be charged simultaneously and the aging impacts on the transformer.

The maximum number of EVs that can be charged concurrently is computed as:

$$N_{\mathrm{EV}} = \frac{S \times \mathrm{PF}}{I \times V \times \phi}, \tag{26}$$

where S is the rated apparent power of the transformer, PF is the power factor, I is the charging current of EV, V is the single-phase voltage, and ϕ is the number of phases. To estimate the number of EVs that can be simultaneously charged at a specific current level, the analysis considers a three-phase charging setup with a per-phase voltage of 230 V. The analysis is based on a 160 kVA distribution transformer operating at a power factor of 0.95, consistent with typical values adopted in the literature [69], [70] (see Table 2). In line with prior transformer aging studies under EV charging scenarios (e.g., [38], [70], [71], [72], [73]), the ambient temperature is fixed at 30 °C, as this value is frequently adopted in the literature and ensures consistency with commonly accepted modelling practices. The number of EVs for a 6 A charging rate is computed as:

TABLE 2: Overview of Technical Specifications of Distribution Transformer (TOT: Top-oil temperature, HST: Hot-spot temperature).

Parameter	Value	Parameter	Value
Rated power	160 kVA	No-load loss	330 W
Primary voltage	11 kV	Full-load loss	2650 W
Secondary voltage	400 V	Ambient Temperature	30 °C
Power factor	0.95	Rated TOT, $\Delta\theta^r_{TO}$	55 °C
Normal Insulation Life	180,000 h	Rated HST, $\Delta\theta^r_{HS}$	25 °C

$$N_{\mathrm{EV}} = \frac{160{,}000 \times 0.95}{6 \times 230 \times 3} = 36.71 \approx 36. \tag{27}$$

In contrast, for a charging current of 15 A, the number reduces to $N_{\mathrm{EV}} = 14$. Notably, many studies assume transformer loading well beyond the rated capacity (e.g., [24], [25]), leading to exaggerated aging factors. In this study, transformer loading is constrained to its rated capacity, reflecting realistic operational limits.

Using the methodology from Section IV, three key transformer performance metrics, hot-spot temperature (HST), aging acceleration factor (F_{AA}), and transformer lifetime, are evaluated for a 160 kVA transformer across charging rates from 6 A to 15 A under full-load conditions. For each charging rate, a randomized selection of EV models (from the eight presented in Section III) is used. The Monte Carlo simulation is iterated one million times to ensure statistical significance; mean values are reported due to negligible confidence intervals.

Figure 7 summarizes the relationship between charging rates and transformer performance indicators. As shown, HST and F_{AA} are positively correlated; higher temperatures accelerate aging, thus reducing transformer lifetime. Additionally, substantial variability is observed due to the random inclusion of high harmonic-emitting EVs in some iterations, explaining the presence of outlier results. It is important to note that the harmonic content at 6 A charging has the highest harmonic content for all EV types. This is also noticeable in the simulation results as EVs with 6A charging leads to highest transformer aging.

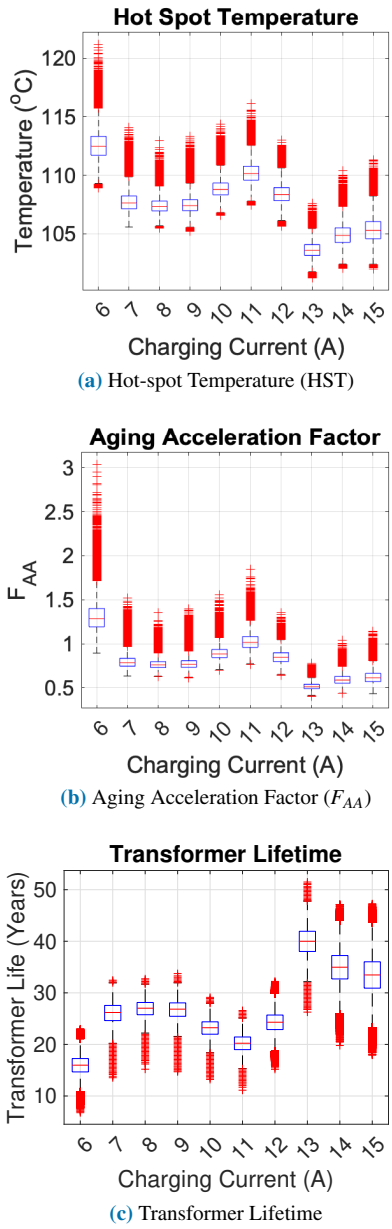
Furthermore, Figure 8 compares transformer aging metrics with and without harmonic distortion. The results highlight the pronounced negative effect of smart charging harmonics on transformer health. For instance, the HST peaks at 6 A, approaching 120°C due to high harmonic content, far above typical thermal limits. As charging rates increase (e.g., 13–15 A), HST stabilizes around 105°C, reflecting reduced harmonic impact. Hence, lower charging currents, while enabling more EVs to charge simultaneously, significantly accelerate transformer aging due to harmonics.

Based on the results presented in Figures 7 and 8, the following *rule-based harmonics-averse charging management* recommendations can be proposed:

- **Preferred charging rate:** 13 A represents an optimal point, minimizing transformer aging while still allowing a reasonable number of EVs to be charged (approximately 16 EVs simultaneously).
- **Low-rate alternative:** If charging at lower rates is necessary (e.g., residential overnight charging), 9 A, 8 A, and 7 A emerges as the least detrimental option. However, longer charging durations will be required.
- **Charging rates to avoid:** 6 A, despite supporting up to 36 EVs, results in excessive transformer aging and should be avoided. Instead, 7 A should be used if low-rate charging is mandatory.

These results confirm that fewer high-current charging sessions (e.g., at 13 A) are more effective in mitigating transformer aging compared to many low-current sessions (e.g., 6–9 A). While lower currents allow more EVs to connect simultaneously, they significantly increase harmonic distortion and prolong transformer thermal stress, thereby accelerating aging.

Rule-based harmonics-averse EV charging management for peak hours is illustrated in Figure 9. As depicted, charging rates are dynamically adjusted based on the number of EVs connected to the system, ensuring that both transformer capacity and harmonic distortion constraints are respected. Each time a new EV requests to charge, equation (26) is evaluated to compute the remaining spare capacity of the transformer.



(a) Hot-spot Temperature (HST)

(b) Aging Acceleration Factor (F_{AA})

(c) Transformer Lifetime

FIGURE 7: Impacts of EV smart charging rates on thermal and aging characteristics of a fully loaded distribution transformer.

The first and most prioritized decision is to allocate a charging rate of 13 A, as this rate minimizes transformer aging while providing sufficiently fast charging. Importantly, since equation (26) evaluated at 13 A also satisfies the capacity constraint for higher rates such as 14 A and 15 A, 13 A is always preferred over these higher rates, making it the optimal upper bound during peak hours.

If the number of EVs continues to increase beyond the point

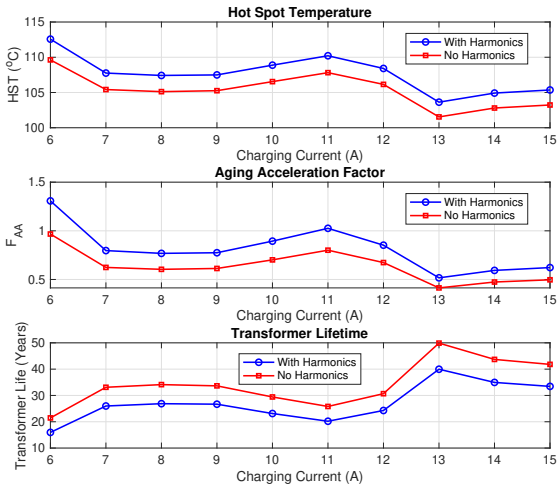


FIGURE 8: Comparison of transformer aging metrics under harmonic-distorted and non-harmonic conditions.

where 13 A charging is not feasible, the system proceeds to lower charging rates in a hierarchical manner. Specifically, 9 A is chosen as the second-best option, given that it offers a better trade-off between aging and capacity compared to intermediate rates such as 10 A, 11 A, or 12 A. As network congestion worsens, the system will further reduce charging rates to 8 A and subsequently to 7 A, with 7 A identified as the lowest acceptable rate based on aging analysis.

Critically, if a new EV connection request would force charging rates below 7 A (e.g., down to 6 A), that request will be denied to avoid excessive transformer aging caused by high harmonic emissions at such low rates. In this way, the rule-based decision tree prevents overloading and mitigates harmonic-related risks without requiring complex, real-time communication or optimization systems. Instead, system operators can rely on a simple yet effective decision-making framework to manage EV charging loads during peak demand periods, balancing infrastructure health and user service needs.

It is important to note that the proposed rule-based EV charging management scheme can be readily extended to scenarios where transformer overloading is permissible. For instance, as illustrated in Figure 10, simulation outcomes are presented for a case in which a 25% transformer overloading margin is tolerated, thereby enabling the connection of additional EVs to the network. However, such an operational mode comes at a significant cost to the asset's health: transformer lifetime is substantially reduced due to elevated hot-spot temperatures exceeding 110 °C. Under these conditions, merely one hour of operation corresponds to an accelerated aging rate that is 15 to 20 times higher compared to the nominal scenario.

Nevertheless, the proposed rule-based framework is inherently flexible and can be adapted to accommodate overloading scenarios with only minor modifications. For example,

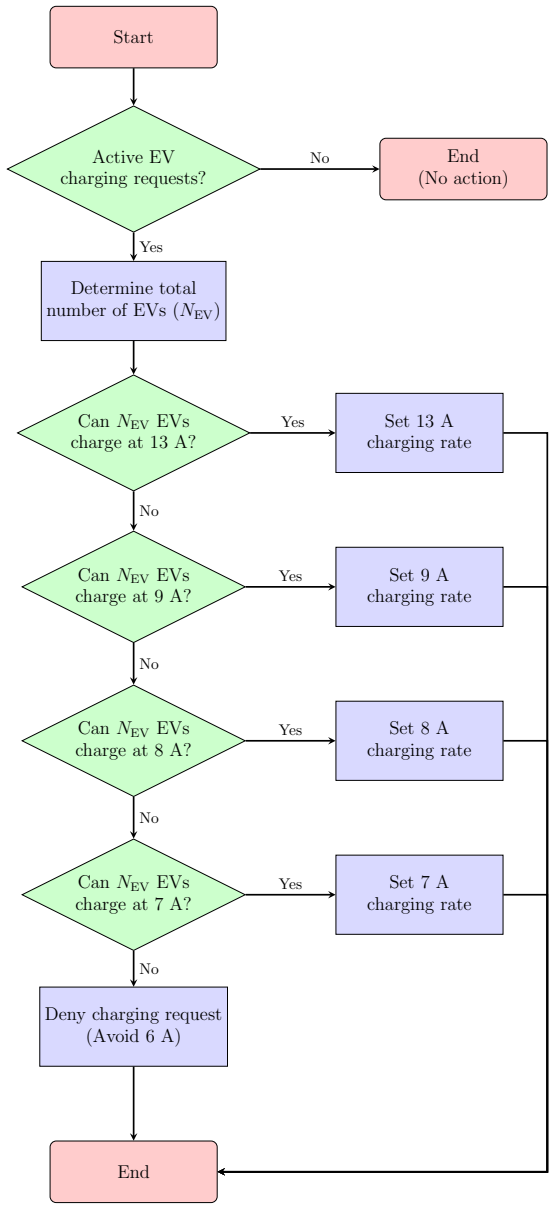
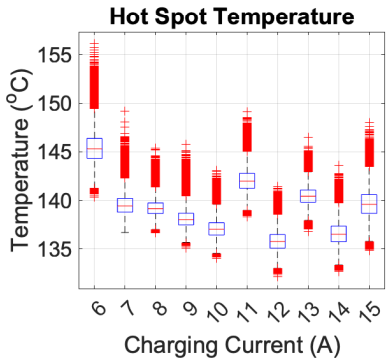


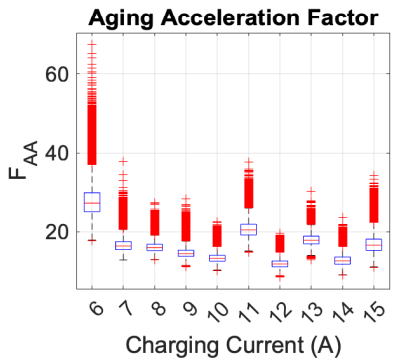
FIGURE 9: Rule-based EV charging management for new connection requests during peak hours.

instead of adopting a fixed charging current of 13 A, a slightly reduced value, such as 12 A, could be employed to mitigate excessive thermal stress. Furthermore, the decision-making logic could be enhanced by incorporating additional charging levels, such as 10 A, to expand the set of operational choices. It should be emphasized, however, that charging at the minimal rate of 6 A remains inadvisable due to its inefficiency and potential to extend charging sessions beyond acceptable durations.

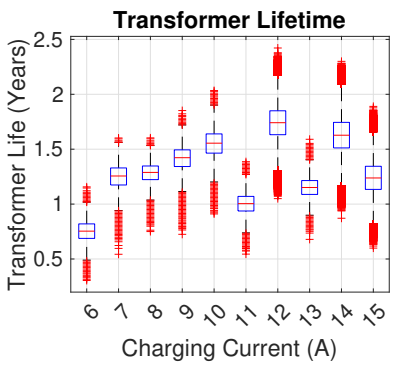
It is important to emphasize that this analysis pertains to heavily loaded networks. In partially loaded transformers, HST is unlikely to exceed 110°C, rendering aging less critical. However, in such cases, THD compliance becomes



(a) Hot-spot Temperature (HST)



(b) Aging Acceleration Factor (F_{AA})



(c) Transformer Lifetime

FIGURE 10: Impacts of EV smart charging rates on transformer thermal and aging characteristics under a 25% overloading scenario.

a primary operational constraint, as mandated by industry standards (e.g., IEEE 519-2014). This distinction underlines the importance of harmonics-aware smart charging strategies, adjusting charging rates based on real-time transformer loading conditions.

B. CASE II: LARGE-SCALE EV PARKING LOTS

To evaluate second case study, a large-scale EV parking lot is considered. It is assumed that a total of 200 EVs request charging service at a workplace charging facility that operates between 6:00 AM and 11:00 PM. Each EV is assumed to remain connected for a minimum parking duration of 8 hours,

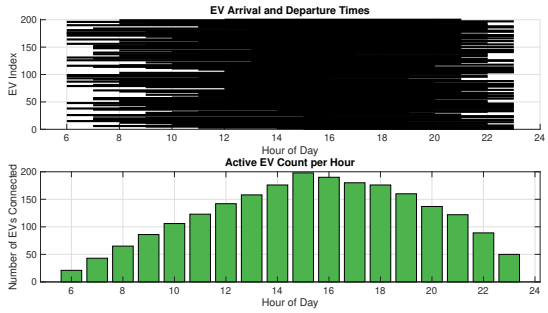


FIGURE 11: Arrival/departure times and the number of active EVs at the charging station.

reflecting typical workday behaviour. Arrival and departure times are generated using a uniform distribution to simulate random user behaviour. It is important to note that Time-of-Use (TOU) electricity pricing is not considered in this study; EV arrival and departure distributions are modelled as uniform to simulate general stochastic user behaviour, and charging decisions are determined centrally via harmonics-aware optimization rather than by individual user responses to TOU signals.

Similarly, charging demands are uniformly distributed between 20 kWh and 30 kWh per phase (equivalent to 60 kWh to 90 kWh in total for three-phase chargers). The EV fleet is composed of $k = 8$ different EV types described in Section III, with equal selection probability of 0.125 for each model.

The charging infrastructure consists of identical three-phase 11 kW AC chargers (corresponding to Level 2 chargers as defined by the SAE J1772 standard), equivalent to 3.68 kW per phase. Following IEEE 519-2014 guidelines, the THD limit is set at 5%. Additionally, the station operator guarantees that at least 90% of each EV's requested energy demand is fulfilled. The allowable per-phase charging power ranges between $c_{\min} = 1.38$ kW (6 A) and $c_{max} = 3.68$ kW (16 A), or 0 A for idle modes. The per-phase capacity constraint (20) is set as $200 \times 0.65 \times 3.68 \approx 480$ kW. Furthermore, the THD of each EV type is modelled using the second-order polynomial fits presented in Table 1.

Moreover, PSO parameters are initialized as follows, swarm size is set to 100 and maximum iteration is set as 500. Inertia weight is set to 0.7, and both cognitive and social coefficients are chosen as 1.4. While the parameter selected greatly affects the computation efficiency and accuracy of results, the parameter setting is made based on commonly made assumptions in the literature [74]. These parameters were finalized by further experimental trials based on computation time and accuracy.

The PSO-based optimization was implemented in MATLAB R2023a and executed on a desktop computer equipped with an Intel Core i7-3770 CPU running at 3.40 GHz and 16 GB of RAM. Under this setup, the unconstrained case completed in approximately 1170 seconds, while the THD-constrained case required around 1190 seconds. These re-

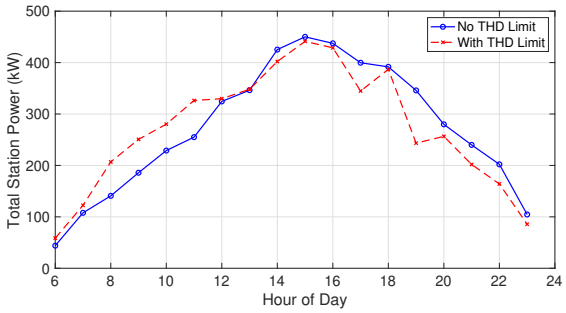


FIGURE 12: Hourly electrical power usage for Optimization Problems I (no THD limit) and II (with THD limit).

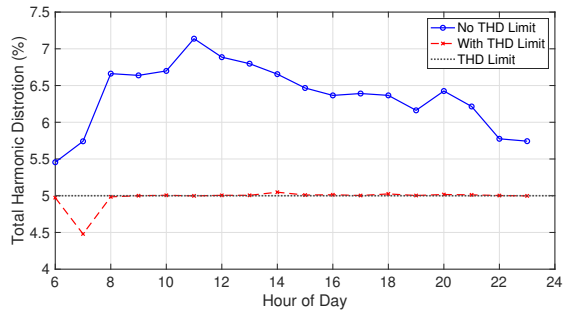


FIGURE 13: Total harmonic distortion (THD) comparison for both optimization cases.

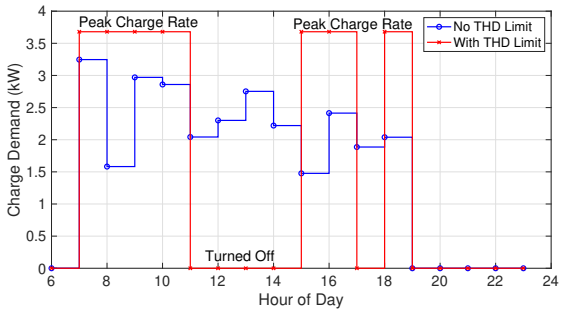


FIGURE 14: Example of THD-aware smart charging for Peugeot e-2008.

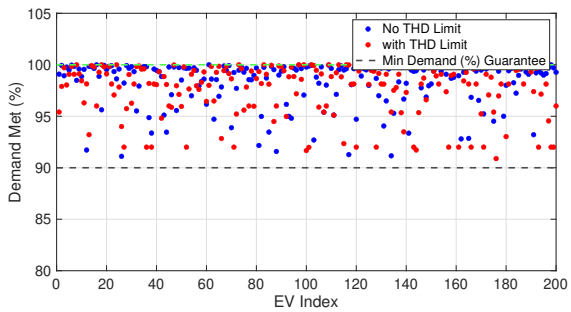


FIGURE 15: Demand completion rates for 200 EVs under both optimization frameworks.

sults demonstrate that the proposed method achieves practical computational efficiency, supporting real-time applications in EV charging infrastructure planning.

Figure 11 presents the arrival and departure times along with the number of active EVs connected throughout the day. The results of the case study are presented in Figures 12 and 13, comparing two optimization frameworks: Optimization Problem I (without THD limits) and Optimization Problem II (with THD limits).

Figure 12 illustrates the hourly power usage of the charging station, which follows the arrival and departure patterns shown in Figure 11. The peak demand occurs around 3:00 PM, with 198 EVs connected and requesting approximately 450 kW per phase. Figure 13 compares the THD resulting from each optimization case. As expected, a regular smart charging algorithm without THD constraints would violate the 5% THD limit during most peak hours, whereas enforcing THD limits in Optimization Problem II successfully keeps harmonic emissions within regulatory bounds. For this case, the THD penalty weight ω is set to one million to strictly enforce harmonic limits, as smaller values lead to violations.

Regarding the mechanism that allows THD-aware smart charging to maintain harmonic levels under control, it is observed that charging at maximum power reduces individual harmonic emissions, as discussed in Figures 2 and 3. Thus, when feasible, EVs are charged at their maximum allowable rates. Additionally, during periods when THD approaches critical limits, some EVs are temporarily turned off, reducing their harmonic contributions to zero. These two strategies form the basis of the harmonics-aware smart charging framework formulated in Optimization Problem II. An example is provided in Figure 14, which illustrates the charging behavior of a Peugeot e-2008. To comply with THD constraints, the vehicle is turned off between 11:00 AM and 3:00 PM and again between 5:00 PM and 6:00 PM. Outside these periods, the vehicle is charged at its rated capacity.

It is noteworthy that peak burst arrivals/departures could increase simultaneous charging events, thereby stressing station capacity and worsening harmonic emissions. In the simulation scenarios, although the charging station experienced a burst of EVs being charged between 2:00 PM and 4:00 PM, the proposed method was still able to maintain THD limits. While the pattern of vehicle arrivals, duration of stay, and charging requests all influence system behaviour, the burst arrival of high-THD-emitting vehicles is a particularly critical factor in THD management. For example, if a large number of Peugeot e-2008 vehicles arrive within a single time slot, the proposed method may no longer be able to satisfy customer demand or maintain THD within acceptable limits.

It is important to analyze how these strategies impact overall energy delivery. Figure 15 presents the demand completion rates for both optimization cases. The results show that the 90% minimum energy requirement is satisfied for all EVs. In the THD-constrained optimization, 97.6% of the total energy requested is delivered, while the unconstrained case achieves 98.9%. The slight reduction in served demand

under THD constraints is an acceptable trade-off to maintain power quality. It is also observed that in both cases, fewer than 1.5% of EVs do not receive their full requested energy. This occurs when the requested demand exceeds what can be delivered within the parking duration (e.g., when a vehicle requests 30 kWh but can only receive up to $3.68 \times 8 = 29.44$ kWh per phase within 8 hours). Nonetheless, the harmonics-aware smart charging framework successfully addresses THD limitations without the need for costly technologies such as active filters.

Finally, it is important to emphasize that as shown in Section III, certain EV models, such as the Peugeot e-2008, introduce disproportionately high harmonic content. The presence of such vehicles can limit the overall reduction of THD achievable at the station level. Therefore, system operators may consider restricting access to high-THD vehicles during peak hours or require that only EVs equipped with harmonics-compliant on-board chargers are permitted to connect.

VII. CONCLUSIONS AND FUTURE WORK

A. DISCUSSION

In this paper, the impacts of smart charging-induced harmonic emissions were assessed under two different scenarios, heavily-loaded and mildly-loaded distribution networks, and mitigation strategies were proposed for each case. In heavily-loaded networks, the primary concern associated with EV smart charging is the additional thermal stress and accelerated aging of distribution transformers caused by increased harmonic content. To assess this impact, a comprehensive Monte Carlo-based impact assessment was conducted, following IEEE C57.91 and IEEE C57.110 guidelines. Eight different EV models were considered, and a 160 kVA distribution transformer was selected as a representative case study.

Key performance indicators, including hottest spot temperature, aging acceleration factor, and estimated transformer lifetime, were calculated under full load conditions. The analysis led to the formulation of a rule-based harmonic-averse EV charging management strategy to be applied during peak demand periods. The results indicated that charging rates could be categorized into two distinct groups, rates higher than 10 A and rates lower than 10 A. Within the higher group, a 13 A charging rate exhibited the least detrimental impact on transformer health and should be prioritized over higher rates such as 14 A and 15 A. Within the lower group, 7 A emerged as the most favorable rate, as it minimizes aging impacts while allowing a larger number of EVs to be charged concurrently. Importantly, the lowest possible charging rate of 6 A was found to be the most harmful in terms of transformer aging and should therefore be avoided. It was further shown that similar methodology can be applied with minor modifications to overloaded transformers.

It is also important to note that transformer aging is driven by the combined contribution of both low- and high-order harmonics. While low-order components such as the 3rd, 5th, and 7th dominate due to their higher amplitudes and associated increase in RMS current, higher-order harmonics additionally elevate losses through frequency-weighted mechanisms in the eddy current and stray loss factors (e.g., h^2 and $h^{0.8}$ terms). This combined effect intensifies winding and core heating, accelerates thermal cycling, and ultimately speeds insulation aging under variable EV charging conditions.

In addition, controlled station loading and harmonics-aware scheduling offer effective mitigation. Prioritizing moderate high-current operation and avoiding very low charging rates reduces harmonic magnitudes and thermal stress, consistent with evidence that controlled utilization (rather than maximum simultaneous charging) yields improved power quality, with diminishing benefits below optimal loading levels. Furthermore, advances in on-board power factor correction (PFC) circuits continue to enhance current shaping and suppress harmonic injections, supporting long-term harmonics mitigation through both smart charging coordination and converter-side design improvements.

To evaluate the case of mildly-loaded networks, a large-scale EV parking lot comprising 200 EVs was analyzed, and a Water Filling methodology embedded into PSO-based smart charging algorithm was proposed to address two optimization problems and to manage THD levels. Optimization Problem I represented a classical smart charging framework designed to satisfy EV energy demand while adhering to charging rate, demand, and parking time constraints. Optimization Problem II built upon the first by incorporating THD constraints that enforced harmonic limits. The proposed solution achieved THD compliance either by increasing EV charging rates to the maximum allowable level, where harmonic emissions are minimized, or by temporarily suspending charging for certain EVs to eliminate their harmonic contributions. In addition, the Water Filling procedure ensures proportional allocation of charging power across EVs under discrete current constraints. Specifically, it prioritizes higher-efficiency charging rates where harmonic emissions are reduced, while strictly avoiding allocations below the minimum 6 A threshold. This prevents infeasible solutions and maintains fairness by distributing energy proportionally among all connected EVs.

The results demonstrated that the proposed harmonics-aware smart charging scheme effectively maintains THD within prescribed limits while still meeting at least 90% of requested charging demands for all users. These findings reinforce that prioritizing moderate high-current charging and operating stations at controlled utilization levels, rather than maximizing simultaneous low-power charging sessions, provides the most effective balance between transformer health, harmonic compliance, and user energy delivery.

It is also worth noting that the PSO-based harmonics-aware strategy explicitly enforces a 5% THD limit, consistent with IEEE 519 and IEC 61000 standards. Maintaining THD within this threshold ensures that harmonic-induced losses remain controlled and that charging operations do not accelerate transformer aging or violate power quality requirements. If future EV penetration levels or charger diversity lead to THD values persistently above this threshold, network reinforce-

ment or dedicated harmonic mitigation equipment (e.g., active filters) would become necessary. Thus, harmonics-aware smart charging not only supports grid reliability and user energy delivery but also helps defer costly network upgrades by maintaining both thermal and harmonic stress within acceptable bounds.

Beyond IEEE 519 and IEC 61000-3-2/-12, which prescribe static harmonic limits based on rated charging current, emerging regulatory frameworks such as the AFIR and UK smart charging legislation emphasize dynamic operational requirements, including flexibility, cybersecurity, and user-centric behavior. These developments indicate that future operator strategies will need to go beyond static harmonic compliance and integrate adaptive, real-time decision-making to balance harmonic reduction, energy delivery, and evolving regulatory mandates.

It is important to emphasize that this study is, to the best of the authors' knowledge, the first to integrate harmonics into smart charging optimization frameworks explicitly. Consequently, direct numerical comparison with prior works is not feasible, as existing studies on EV charging optimization or transformer aging have not incorporated harmonics-aware models. Instead, this study extends the scope of earlier works by demonstrating how harmonics-aware management can provide new theoretical insights into EV–grid interactions and practical guidance for operators facing both power quality regulations and asset protection requirements.

B. LIMITATIONS OF RESEARCH

This study has several limitations that should be acknowledged. First, transformer aging assessments were conducted under a single ambient temperature condition. However, as reported in [24], higher ambient temperatures can significantly elevate transformer hottest spot temperature and aging acceleration factors, potentially exacerbating the aging impact. Second, while a 160 kVA transformer was considered in this study, transformers with lower ratings, such as 25 kVA units, would experience more severe aging effects with fewer EVs connected. In such cases, the hottest spot temperature would increase more rapidly, and aging acceleration would grow exponentially.

Regarding the mildly-loaded network case, the THD-aware smart charging framework relies on second-order polynomial fits for individual EV THD modelling and employs arithmetic mean approximations for aggregating multiple EV harmonic contributions. Although this method allows for efficient optimization, it introduces a conservative bias, leading to a slight over-provisioning of charging schedules compared to exact vector-based THD summation. As a result, while THD limits are respected, some network capacity may remain underutilized. This conservative assumption may slightly overestimate harmonic impacts in large EV fleets where partial cancellation effects can occur, and thus represents an intentional trade-off between accuracy and computational tractability.

Finally, the harmonic models were developed based on measurements from eight EV models, representing a broad range of on-board charger designs. Although these models capture a wide variety of harmonic profiles, future studies could benefit from including additional EV types to further enrich the model library and improve the generalizability of the results.

C. FUTURE WORK

Future research will address the aforementioned limitations and explore several key directions:

- The harmonics-aware smart charging framework will be expanded to include a broader set of EV models and real-world harmonic imbalance scenarios. While the present analysis assumes balanced three-phase conditions, practical charging environments frequently involve mixed single- and three-phase EVs, which can introduce phase imbalance, neutral current stress, and unbalanced harmonic propagation. Future work will therefore incorporate aggregate and individual harmonic unbalance indices based on Fortescue symmetrical component theory [75] to explicitly model harmonic imbalance effects and more accurately assess transformer aging and power-quality performance under heterogeneous charging conditions.
- The arithmetic mean approximation for upper-bound THD estimation will be refined by developing meta-models, such as response surface methodologies, that may better capture vector summation behavior while maintaining computational tractability. This will help to narrow the gap between approximate and actual THD levels.
- Transformer aging assessments will be extended to include varying ambient temperatures and broader climatic conditions to more accurately reflect real-world operational settings. Moreover, aging impacts will be analyzed for a wider range of transformer capacities, including smaller distribution transformers, to assess scalability of results.
- The EV harmonics database will be expanded through additional field measurements and laboratory tests, including finer-granularity charging current levels (e.g., decimal values), which will enable the use of vector-based harmonic summation for more accurate THD calculations and aging models.
- The EV harmonics database will be expanded through additional field measurements and laboratory tests, including finer-granularity charging current levels (e.g., decimal values). This will enable the use of fleet-aware vector-based harmonic summation methods, reducing the conservativeness of the current arithmetic mean approximation, which may slightly overestimate THD in large EV fleets due to ignoring cancellation effects. These improvements will support more accurate THD calculations and transformer aging models.
- In this study, uniform distributions were used for EV arrival/departure times and energy demands to represent general stochastic behavior without introducing

temporal bias. However, real-world EV usage patterns often exhibit clustering, such as morning and evening arrival peaks, and diverse user requirements ranging from urgent to flexible charging. Deviations from uniformity, such as concentrated arrivals during peak hours, may result in higher simultaneity in charging, potentially increasing station-level power demand and aggregate harmonic distortion. Future work will explore the robustness of the proposed harmonics-aware charging framework under more realistic, non-uniform EV arrival models, including probabilistic or empirically derived distributions based on real-world datasets.

- Future work could also incorporate user behavior in response to Time-of-Use (TOU) electricity pricing schemes. While this study assumes uniformly distributed arrival and departure times to reflect stochastic user patterns, integrating TOU-driven decision-making could further enhance the realism of EV charging models and explore potential synergies between cost-driven user strategies and harmonics-aware optimization.
- While this study employed a simple rule-based framework (e.g., prioritizing 13 A charging to mitigate transformer aging and harmonics), it is intended as an interim, practical solution to facilitate the transition toward broader EV adoption and grid harmonization. Long-term mitigation, however, will depend on the development of on-board chargers designed in accordance with emerging industry standards to inherently minimize harmonic emissions. Future research could also explore adaptive or learning-based charging strategies that better balance user convenience, harmonic reduction, and transformer health under diverse and dynamic operating conditions, thereby improving robustness beyond fixed rule sets.
- Future studies should incorporate fairness metrics and explicitly model user delays in charging schedules to better quantify user satisfaction. These metrics would complement the proportional allocation inherent in the Water Filling framework, enabling a more comprehensive assessment of harmonics-aware charging strategies.

By addressing these research directions, utilities and policymakers will be better equipped to manage the dual challenges of high EV penetration and grid asset protection, thereby supporting both electrification goals and the long-term reliability of distribution networks.

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